Quarry: Test Strategy and QA

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Document	What is tested, how, and what is still unverified
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1. Quality goals

1. No regression in the core flow: describe an ICP, get leads, export.
2. No cross-tenant access, no unsafe outbound fetch, no unverified payment credit.
3. Failures are visible to the user with a reason, never silent.
4. Every external service can be faked, so tests are fast, deterministic and free.

2. Test pyramid

Level	Tooling	Count	What it covers
Backend unit and integration	pytest	478 collected	Pipeline, jobs, scoring, exports, LLM failure paths, auth and CSRF, ownership, SSRF, migrations, billing, outreach, watch, waterfall, compliance
Frontend component and page	Vitest and Testing Library	375 passing in 39 files	Pages, dialogs, forms, states, accessibility behaviors
Static analysis	ruff, eslint	Every push	Style and common defects
Build	`vite build`	Every push	The production bundle compiles
Browser QA	Manual and scripted passes	Two passes recorded	Real flows in a browser; findings logged in `TODOS.md`
End-to-end against live services	None automated	n/a	**Gap**, see section 6

3. How the suite is built

- `tests/conftest.py` seeds throwaway environment variables, so no real credentials are ever used.
- The `db` fixture gives every test a fresh SQLite file.
- **All external calls are mocked or faked**: LLM, Serper, page fetches, SMTP, IMAP, payment providers, CRM, enrichment providers. Each integration sits behind a port (an HTTP client protocol, a `PaymentProvider`, a `CrmConnector`), and tests inject a fake.
- The frontend mocks the API module and asserts on roles, labels and visible states, not implementation details.
- CI (`.github/workflows/ci.yml`) runs on pushes and pull requests: ruff, pytest, eslint, vitest and the web build.

4. Coverage map

Area	Evidence (test modules)
Authentication, cookies, CSRF, refresh rotation	`test_auth_cookies.py`, `tests/db/test_identity.py`
Ownership and request hardening	`test_flow_hardening.py`, `tests/db/test_hardening.py`, `test_dependencies.py`
Pipeline end to end	`test_pipeline.py`
LLM failure handling and retry	`test_llm_failures.py`
Scoring and exports	`test_scoring_and_export.py`
Durable jobs: retry, backoff, cancel, recovery	`test_jobs.py`
SSRF guard, DNS pinning, redirects, MX	`test_url_guard.py`, `_pinning`, `_mx`, `_careers`, `_tech_stack`, `_other_fetchers`
Database layer and migrations	`tests/db/*` including `test_migrations.py`
Email verification statuses, provenance, region, cost and usage, compliance	`tests/phase2/*`
Outreach logic, footer, flow, mailbox connection, watch, waterfall, integrations	`tests/p3/*`
Teams, billing, top-ups, CRM, invite email, background work	`tests/growth/*`
Deployment environment	`test_deployment_env.py`
Frontend pages and dialogs	`*.test.jsx` beside each page, including mailbox test connection and regression tests

5. Defect history and what it taught

Browser QA passes found issues that mocked tests did not. Each was fixed and, where possible, covered by a test.

Finding	Lesson
A hot-reloading dev server created a table with another phase's columns mid-merge and Billing returned HTTP 500	`create_all` never alters tables: added the versioned migration runner and a unique-table-name test
A mailbox with an unreachable host saved without any check	Added "Test connection" before saving
Team invite blocked the request about 4 seconds while SMTP was tried	Send in the background and return an `email_status`
Error toasts covered the primary button of the form above	Toasts moved to the top
Dialog action buttons scrolled out of view on short screens	Sticky dialog footer

6. Known gaps (not yet verified)

Gap	Risk	Plan
Real-service coverage: the full pipeline with a live Serper key, request details and the lead drawer with real leads, email verification providers, outbound SMTP and IMAP, Stripe, Razorpay, HubSpot, the live cost estimate and sample run	Mocks can drift from real provider behavior	A scripted smoke run per integration before hosting; record results
Shared requests and lists with two members in one workspace	The “Shared with” sections were only checked with one member	Two-account test, automated if possible
LLM extraction accuracy	Hallucinated or wrong contacts	Build a labelled set of 100 to 200 leads; track precision and bounce rate
Run duration and cost under load	Unknown capacity	Time real runs; record p50 and p95
No browser end-to-end suite	UI regressions between releases	Add Playwright smoke tests for sign-up, launch, results, export
No load or soak test	Single-writer behavior under concurrency	A small concurrent-launch test
No external security test or dependency scan	Unknown vulnerabilities	`pip-audit`, `npm audit` in CI; pen test before hosting
Accessibility audit	Contrast of accents and status chips unchecked	Run an automated and manual audit across the seven palettes

7. Release gate

A change may merge when:

1. `ruff` and `eslint` are clean.
2. `pytest` and `vitest` pass in CI.
3. `vite build` succeeds.
4. A schema change ships with a migration and a migration test.
5. A bug fix ships with a regression test that fails without the fix.
6. User-facing changes have been checked in a browser at desktop and phone width.

8. Manual test checklist (pre-release)

Flow	Steps	Expected
Sign-up and login	Register, log out, log in, Google sign-in	Cookies set; protected pages load; logout clears them
Launch a run	Wizard, estimate, launch	202, live progress, results appear, status Completed
Failure visible	Use a bad Serper key	Request Failed with a reason; partial leads kept
Enrichment retry	Force an LLM failure, press Retry	Lead updates in place
Export	CSV, XLSX, PDF from a request and a list	Files open; source-notice line present; suppressed contacts absent
Suppression	Add an address, enroll and export	Skipped at send and export
Outreach	Connect and test a mailbox, preflight, sequence, enroll, reply	Sends inside the window and cap; reply stops the contact
Unsubscribe	Open the footer link	Suppression entry and audit record created
Billing	Top up on the mock provider; replay a webhook	Credited once
Team	Invite, accept, change role, share a list	Role limits enforced; teammate sees shared items
CRM	Connect HubSpot, push a list	202 job; per-lead sync status updates
Erasure	Erase a person	Removed everywhere; audit entry written
Responsive	1280x720 and phone width	No clipped actions; dialogs keep their footer visible